Week 1 Problem Set: Limits from Graphs and Tables

Calculus Honors · Module 1: The Idea of a Limit · Lessons 1.01 and 1.02

Name: _____ Date: _____

> Show all of your work. A correct answer with no reasoning earns partial credit at most. Keep your writing neat and vertical, line up your equal signs, and label your final answers. You may check your estimates with a graphing tool, but the reasoning must be your own.

PART A · ESTIMATING LIMITS FROM A TABLE

1. Complete the table for $f(x) = \dfrac{x^2 - 9}{x - 3}$, then estimate $\lim_{x \to 3} f(x)$. Explain in one sentence why the limit exists even though $f(3)$ is undefined.

x	2.9	2.99	2.999	3.001	3.01	3.1
$f(x)$						

2. Build your own two-sided table (three values on each side) to estimate $\lim_{x \to 0} \dfrac{\sqrt{x+1} - 1}{x}$. State your estimate to three decimal places.

PART B · THE LIMIT LAWS AND DIRECT SUBSTITUTION

3. Evaluate each limit by direct substitution and name why substitution is allowed. (a) $\lim_{x \to 2} (3x^2 - 5x + 1)$ (b) $\lim_{x \to -1} \dfrac{x + 4}{x^2 + 3}$ (c) $\lim_{x \to 4} \sqrt{2x + 1}$

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4. Given $\lim_{x \to a} f(x) = 6$ and $\lim_{x \to a} g(x) = -2$, find each limit and state the law you used. (a) $\lim [f(x) + g(x)]$ (b) $\lim [f(x)\, g(x)]$ (c) $\lim \dfrac{f(x)}{g(x)}$

PART C · READING A GRAPH

5. The graph of $y = g(x)$ passes through a straight segment with an open circle at $(1, 3)$ and a separate solid dot at $(1, 5)$. Determine (a) $\lim_{x \to 1} g(x)$ and (b) $g(1)$. Explain why they are not the same, and which one the limit reports.

PART D · EXPLAIN YOUR THINKING

6. Convince a classmate, in three or four sentences, why the value $f(a)$ is irrelevant to $\lim_{x \to a} f(x)$. Use one of the examples above as evidence.